

Level I**(Problems Based on Fundamentals)**

- Find the interval of the monotonicity of the function $f(x) = 2x^3 - 12x^2 + 18x + 5$.
- Find the interval of the monotonicity of the function $f(x) = 5 + 36x + 3x^2 - 2x^3$.
- Find the interval of the monotonicity of the function $f(x) = (x - 1)^3(x - 2)^2$.
- Find the interval of the monotonicity of the function $f(x) = 2x^3 - 3x^2 + 6x + 10$.
- Find the interval of the monotonicity of the function $f(x) = 2x^3 + 3x^2 + 12x + 20$.
- Find the interval of the monotonicity of the function $f(x) = \frac{x}{2} + \frac{2}{x}$.
- Find the interval of the monotonicity of the function $f(x) = 5x^{3/2} - 3x^{5/2}$, $x > 0$.
- Find the interval of the monotonicity of the function $f(x) = \log(x + \sqrt{1 + x^2})$.
- Find the interval of the monotonicity of the function $f(x) = \frac{x}{\log x}$.
- Find the interval of the monotonicity of the function $f(x) = x - \cot^{-1} x - \log(x + \sqrt{x^2 + 1})$.
- Find the least value of m for which the function $f(x) = -x^2 + mx + 1$ is strictly increasing in $[1, 2]$.
- For what values of b , the function $f(x) = \sin x - bx + c$ is strictly decreasing for all x in R .
- Find all possible values of ' a ' for which the function $f(x) = e^{2x} - (a + 1)e^x + 2x$ is strictly increasing for all x in R .
- For what values of a is the function $f(x) = \left(\frac{a^2 - 1}{3}\right)x^3 + (a - 1)x^2 + 2x + 1$ strictly increasing?
- For what values of a , the function $f(x) = (a + 2)x^3 - 3ax^2 + 9ax - 1$ is strictly decreasing for all x in R .

Critical Points

- Find the critical points of $f(x) = \frac{e^x}{x - 1}$.
- Find the critical points of $f(x) = \frac{5x^2 - 18x + 45}{x^2 - 9}$.
- Find the critical points of the function $f(x) = x^{4/5}(x - 4)^2$.
- Find the critical points of the function $f(x) = x + \cos^{-1} x + 1$.
- Find the critical points of the function $f(x) = \sqrt{x^2 - 6x + 15}$.

Increasing and Decreasing Functions

- Find the interval of increasing and decreasing of a function $f(x) = 2x^2 - \ln|x|$.
- Find the intervals for the function $f(x) = \frac{|x - 1|}{x^2}$ increasing and decreasing.
- Find the intervals for the function $f(x) = x^2 e^{\frac{-x^2}{a^2}}$, $a > 0$ is increases. Isolation points.
- Show that the equation $x^3 = 3x + 1$ has a real root in $[-1, 1]$.
- Show that the equation $e^x = 1 + x + \frac{x^2}{2}$ has a real root in $[-1, 1]$.

Algebra of Monotonic Functions

- Find the interval where the function $f(x) = \tan^{-1}$ is strictly increasing.
- Find the interval in which $f(x) = \tan^{-1}(\log_{1/3} x)$ strictly decreasing.
- Find the interval in which $f(x) = \cot^{-1}(\log_4 x)$ strictly decreasing.
- Find the interval in which $f(x) = \cot^{-1}(\log_{1/10} x)$ strictly increasing.
- Find the interval of the monotonicity of the function $f(x) = \sqrt{3x - x^2}$.
- Find the interval of the monotonicity of the function $f(x) = \tan^{-1}(\sin x + \cos x)$ in $(0, 2\pi)$.

32. Find the interval of the monotonicity of the function

$$f(x) = \log\left(\frac{\log x}{x}\right).$$

33. Find the interval in which the function

$$f(x) = \sin(\log x) + \cos(\log x) \text{ is decreases.}$$

34. Find the interval of the monotonicity of the function

$$f(x) = \log_e(\cos x) \text{ for all } x \in (0, \pi).$$

35. Find the interval of the monotonicity of the function

$$f(x) = \sin(\sin x) + \cos(\sin x) \text{ in } (0, \pi).$$

Inequality

36. Prove the inequality, $\log(1+x) > x - \frac{x^2}{2}$ for all x in R^+

37. Prove the inequality $\log(1+x) > \frac{x}{1+x}$ for $x > 0$

38. Prove that $(e^x - 1) > (1+x)\log(1+x)$, if $x > 0$

39. Prove that $2x \tan^{-1} > \log(1+x^2)$ for all x in R^+ .

40. Prove that $1+x \log(x + \sqrt{x^2+1}) \geq \sqrt{1+x^2}$ for all $x \geq 0$.

41. Prove that $\cos(\sin x) > \sin(\cos x)$, if $x \in (0, \frac{\pi}{2})$.

42. Find the smallest positive constant B such that $x \leq Bx^2$ for all $x > 0$.

43. If $x^2 + \frac{b}{x} \geq c$, $\forall x \in R^+$, where a, b, c are +ve constants, prove that $27ab^2 \geq 4c^3$

Concavity

44. Find the interval of the concavity of the function $f(x) = x^5 + 5x - 6$.

45. Find the interval of the concavity for the function $f(x) = x^4 - 5x^3 - 15x^2 + 30$.

46. Find the interval of the concavity for the function $f(x) = (\sin x + \cos x)e^x$ in $(0, 2\pi)$

47. Show that the curve $y = f(x) = Ax^2 + Bx + c$ is concave up if $A > 0$ and concave down if $A < 0$.

Point of Inflection

48. Find the inflection point of the function

$$f(x) = x^4 - 4x^3 + x - 10$$

49. Find the point of inflection of the curve

$$y = f(x) = (x-2)^{2/3} + 10$$

50. Find the point of inflections of the curve

$$f(x) = x^4 - 6x^3 + 12x^2 - 8x + 3$$

51. Find the point of inflection of the curve

$$y = f(x) = x^2 - \frac{1}{6x^3}$$

52. Find the inflection point of the curve $y = f(x) = e^{-x^2}$

Hints & Solutions

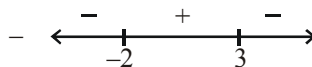
Level - 1

1. Given, $f(x) = 2x^3 - 12x^2 + 18x + 5$
 $= 6x^2 - 24x + 18$
 $= 6(x^2 - 4x + 3)$
 $= 6(x^2 - 4x + 3)$
 $= 6(x - 1)(x - 3)$



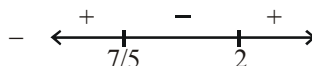
By the sign scheme, $f(x)$ is strictly increases in $(-\infty, 1) \cup (3, \infty)$ and strictly decreases in $(1, 3)$

2. Given $f(x) = 5 + 36x + 3x^2 - 2x^3$
 $\Rightarrow f'(x) = 36 + 6x - 6x^2$
 $= -6(x^2 - x - 6)$
 $= -6(x - 3)(x + 2)$



By the sign scheme, we can say that, $f(x)$ is strictly increases in $(-2, 3)$ and strictly decreases in $(-\infty, -2) \cup (3, \infty)$

3. Given $f(x) = (x - 1)^3 (x - 2)^2$
 $\Rightarrow f'(x) = 3(x - 1)^2 (x - 2)^2 + 2(x - 1)^3 (x - 2)$
 $= (x - 1)^2 (x - 2) (3(x - 2) + 2(x - 1))$
 $= (x - 1)^2 (x - 2) (5x - 7)$



By the sign scheme, we can say that, $f(x)$ is strictly increase in $\left(-\infty, \frac{7}{5}\right) \cup (2, \infty)$ and strictly decreases in $\left(\frac{7}{5}, 2\right)$

4. Given $f(x) = 2x^3 - 3x^2 + 6x + 10$
 $\Rightarrow f'(x) = 6x^2 - 6x + 6$
 $= 6(x^2 - x + 2) > 0$, for all x in $\mathbb{R}$

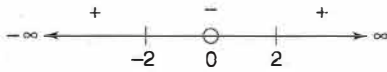
5. Given $f(x) = 2x^3 + 3x^2 + 12x + 20$
 $f'(x) = 6x^2 + 6x + 2$
 $= 6(x^2 + x + 2) > 0$ for all x in $\mathbb{R}$.

Thus, $f(x)$ is strictly increases in $(-\infty, \infty)$

6. Given $f(x) = \frac{x}{2} + \frac{2}{x}$

$$\Rightarrow f'(x) = \frac{1}{2} - \frac{2}{x^2} = \frac{x^2 - 4}{2x^2}$$

$$= \frac{(x-2)(x+2)}{2x^2}$$

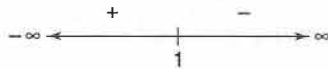


By the sign scheme, we can say that, $f(x)$ is strictly increases in $(-\infty, -2) \cup (2, \infty)$ and strictly decreases in $(-2, 0) \cup (0, 2)$

7. Given $f(x) = 5x^{3/2} - 3x^{5/2}, x > 0$

$$\Rightarrow f'(x) = 15x^{1/2} - 15x^{3/2}$$

$$= 15\sqrt{x}(1-x) - 15\sqrt{x}(x-1)$$



By the sign scheme, we can say that, $f(x)$ is strictly increases in $(0, 1)$ and strictly decreases in $(1, \infty)$

8. Given $f(x) = \log(x + \sqrt{1+x^2})$

$$\Rightarrow f'(x) = \frac{1}{(x + \sqrt{x^2 + 1})} \left(1 + \frac{1 \times 2x}{2\sqrt{x^2 + 1}} \right)$$

$$\Rightarrow f'(x) = \frac{1}{(x + \sqrt{x^2 + 1})} \left(1 + \frac{x}{\sqrt{x^2 + 1}} \right)$$

$$= \frac{1}{(x + \sqrt{x^2 + 1})} \left(\frac{\sqrt{x^2 + 1} + x}{\sqrt{x^2 + 1}} \right)$$

$$= \frac{1}{\sqrt{x^2 + 1}} > 0 \quad \forall x \in \mathbb{R}$$

Thus, $f(x)$ is strictly increases in $(-\infty, \infty)$

9. Given $f(x) = \frac{x}{\log x}$

$$\Rightarrow f'(x) = \frac{\log x - 1}{(\log x)^2}$$



By the sign scheme, we can say that, $f(x)$ is strictly increases in (e, ∞) and strictly decreases in $(0, e)$.

10. Given $f(x) = \cot^{-1} x - \log(x + \sqrt{x^2 + 1})$

$$\Rightarrow f'(x) = 1 + \frac{1}{1+x^2} - \frac{1}{(x + \sqrt{x^2 + 1})} \times \frac{(x + \sqrt{x^2 + 1})}{\sqrt{x^2 + 1}}$$

$$\Rightarrow f'(x) = 1 + \frac{1}{1+x^2} - \frac{1}{\sqrt{x^2 + 1}} > 0 \quad \forall x \in \mathbb{R}$$

Thus, $f(x)$ is strictly increases in $(-\infty, \infty)$

11. Given $f(x) = -x^2 + mx + 1$

$$\Rightarrow f'(x) = -2x + m$$

Since f is strictly increasing, so $f'(x) > 0$

$$\Rightarrow -2x + m > 0$$

$$\Rightarrow m > 2x$$

$$\Rightarrow m > 2, \quad \forall x \in [1, 2]$$

Hence, the least value of m is 2.

12. Given $f(x) = \sin x - bx + c$

$$\Rightarrow f'(x) = \cos x - b$$

Since f is strictly decreasing, so $f' < 0$

$$\Rightarrow \cos x - b < 0$$

$$\Rightarrow b > \cos x$$

$$\Rightarrow b > 1$$

Hence, $b \in (1, \infty)$

13. Given $f(x) = e^{2x} - (a+1)e^x + 2x$

$$\Rightarrow f'(x) = 2e^{2x} - (a+1)e^x + 2$$

$$= 2(e^x)^2 - (a+1)e^x + 2$$

Since f is strictly increasing, so $f'(x) > 0$

$$\Rightarrow 2(e^x)^2 - (a+1)e^x + 2 > 0$$

$$\Rightarrow 2e^x - (a+1)e^x + \frac{2}{e^x} > 0$$

$$\Rightarrow 2\left(e^x + \frac{1}{e^x}\right) - (a+1) > 0$$

$$\Rightarrow (a+1) < 2\left(e^x + \frac{1}{e^x}\right)$$

$$\Rightarrow (a+1) < 2.2 = 4$$

$$\Rightarrow a < 3$$

Hence, the value of a is $\in (-\infty, 3)$

14. Given $f(x) = \left(\frac{a^2-1}{3}\right)x^3 + (a-1)x^2 + 2x + 1$

$$\Rightarrow f'(x) = 3\left(\frac{a^2-1}{3}\right)x^2 + 2(a-1)x + 2$$

$$\Rightarrow f'(x) = (a^2 - 1)x^2 + 2(a - 1)x + 2$$

Since f is strictly increasing, so $f'(x) > 0$

$$\Rightarrow (a^2 - 1)x^2 + 2(a - 1)x + 2 > 0$$

$$\Rightarrow (a^2 - 1) > 0 \text{ \& } 4(a - 1)^2 - 8(a^2 - 1) < 0$$

$$\Rightarrow a^2 > 1 \text{ \& } (a - 1)^2 - 2(a^2 - 1) < 0$$

$$\Rightarrow (a + 1)(a - 1) > 0 \text{ \& } (a + 3)(a - 1) > 0$$

$$\Rightarrow a \in (-\infty, -1) \cup (1, \infty)$$

$$\text{and } a \in (-\infty, -3) \cup (1, \infty)$$

Hence, the values of a are $a \in (-\infty, -3) \cup (1, \infty)$

$$15. \text{ Given } f(x) = (a + 2)x^3 - 3ax^2 + 9ax - 1$$

$$\Rightarrow f'(x) = 3(a + 2)x^2 - 6ax + 9a$$

Since f is strictly decreasing for all x in R , so

$$f'(x) < 0$$

$$\Rightarrow 3(a + 2)x^2 - 6ax + 9a < 0$$

$$\Rightarrow (a + 2)x^2 - 2ax + 3a < 0$$

$$\text{Thus, } (a + 2) < 0 \text{ and } 4a^2 - 12a(a + 2) < 0$$

$$\Rightarrow a < -2 \text{ and } a^2 - 3a(a + 2) < 0$$

$$\Rightarrow a < -2 \text{ \& } a(a + 3) > 0$$

$$\Rightarrow a \in (-\infty, -2) \text{ \& } a \in (-\infty, -3) \cup (0, \infty)$$

$$\text{Thus, } a \in (-\infty, -3)$$

$$16. \text{ Given } f(x) = \frac{e^x}{x - 1}$$

$$\Rightarrow f'(x) = \frac{(x - 1)e^x - e^x \cdot 1}{(x - 1)^2} = \frac{e^x(x - 2)}{(x - 1)^2}$$

$$\text{Also, } D_f = R - \{1\}$$

Since $x = 1$ is not an interior point in the domain of f , so, $x = 1$ is not a critical point of f .

Thus, the critical point of f is $x = 2$.

$$17. \text{ Given } f(x) = \frac{5x^2 - 18x + 45}{x^2 - 9}$$

$$\Rightarrow f'(x) = \frac{(x^2 - 9)(10x - 18) - (5x^2 - 18x + 45) \cdot 2x}{(x^2 - 9)^2}$$

$$\Rightarrow f'(x) = \frac{18(x^2 - 10 + 9)}{(x^2 - 9)^2}$$

$$\Rightarrow f'(x) = \frac{18(x - 1)(x - 9)}{(x^2 - 9)^2}$$

$$\text{Also, } D_f = R - \{-3, 3\}$$

Since $x = -3, 3$ are not an interior point of the domain of f , so $x = -3, 3$ are not the critical point of f .

Thus, the critical points of f are $x = 1$ and 9 .

$$18. \text{ Given } f(x) = x^{4/5}(x - 4)^2$$

$$\Rightarrow f'(x) = 2x^{4/5}(x - 4) + \frac{4}{5x^{1/5}}(x - 4)^2$$

$$\Rightarrow f'(x) = \frac{10x(x - 4) + 4(x - 4)^2}{5x^{1/5}}$$

$$\Rightarrow f'(x) = \frac{(x - 4)(10x + 4x - 16)}{5x^{1/5}}$$

$$\Rightarrow f'(x) = \frac{2(x - 4)(7x - 8)}{5x^{1/5}}$$

$$\text{Also, } D_f = R$$

Thus, the critical points of f are $x = 0, \frac{8}{7}, 4$

$$19. \text{ Given } f(x) = x + \cos^{-1}x + 1$$

$$\Rightarrow f'(x) = \frac{1 - 1}{\sqrt{1 - x^2}}$$

$$\text{Now, } f'(x) = 0 \Rightarrow 1 - \frac{1}{\sqrt{1 - x^2}} = 0$$

$$\Rightarrow \frac{1}{\sqrt{1 - x^2}} = 1$$

$$\Rightarrow x = 0.$$

$$\text{Also, } D_f = [-1, 1]$$

Thus, the critical points of ' f ' is $x = 0$.

$$20. \text{ Given } f(x) = \sqrt{x^2 - 6x + 15}$$

$$\Rightarrow f'(x) = \frac{1}{2\sqrt{x^2 - 6x + 15}} \times (2x - 6)$$

$$\Rightarrow f'(x) = \frac{1}{2\sqrt{x^2 - 6x + 15}} \times 2(x - 3)$$

$$\Rightarrow f'(x) = \frac{(x - 3)}{\sqrt{x^2 - 6x + 15}}$$

$$\text{Also } D_f = R$$

Thus, the critical points of f is $x = 3$.

$$21. \text{ Given } f(x) = 2x^2 - \ln|x|$$

$$\Rightarrow f'(x) = 4x - \frac{1}{x} = \frac{4x^2 - 1}{x}$$

$$= \frac{(2x - 1)(2x + 1)}{x}$$

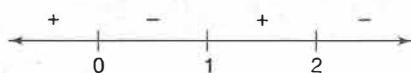


By the sign scheme for $f'(x)$, we have, $f(x)$ is increasing in $\left[-\frac{1}{2}, 0\right) \cup \left[\frac{1}{2}, \infty\right)$ and decreasing in $\left(-\infty, -\frac{1}{2}\right) \cup \left(0, \frac{1}{2}\right]$.

22. Given $f(x) = |x - 1|/x^2$

$$\Rightarrow f(x) = \begin{cases} \frac{x-1}{x^2} = \frac{1}{x} - \frac{1}{x^2} & : x \geq 1 \\ \frac{1-x}{x^2} = \frac{1}{x^2} - \frac{1}{x} & : x < 1 \end{cases}$$

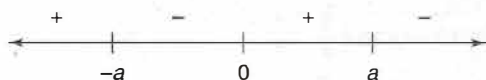
$$\Rightarrow f'(x) = \begin{cases} -\frac{1}{x^2} + \frac{2}{x^3} = \frac{2-x}{x^3} & : x \geq 1 \\ \frac{2}{x^3} - \frac{1}{x^2} = \frac{x-2}{x^3} & : x < 1 \end{cases}$$



By the sign scheme for the function $f'(x)$, we have $f(x)$ is increasing in $(-\infty, 0) \cup [1, 2]$ and decreases in $(0, 1) \cup [2, \infty)$.

23. Given $f(x) = x^2 e^{-x^2/a^2}$, $a > 0$

$$\begin{aligned} \Rightarrow f'(x) &= 2x e^{-x^2/a^2} + x^2 \cdot e^{-x^2/a^2} \times \left(-\frac{2x}{a^2}\right) \\ &= 2x e^{-x^2/a^2} \left(1 - \frac{x^2}{a^2}\right) \\ &= -2x e^{-x^2/a^2} \left(\frac{(x-a)(x+a)}{a^2}\right) \\ &= \left(-\frac{2}{a^2}\right) e^{-x^2/a^2} \times x(x-a)(x+a) \end{aligned}$$



Now, by the sign scheme for the function $f'(x)$, we have $f(x)$ is increases in $(-\infty, -a] \cup [0, a]$

24. Given $x^3 = 3x + 1$

Let $f(x) = x^3 - 3x - 1$

$$\Rightarrow f'(x) = 3x^2 - 3 = 3(x^2 - 1)$$

$$\Rightarrow f'(x) = 3(x^2 - 1) < 0, \forall x \in (-1, 1)$$

Thus $f(x)$ is strictly decreases in $(-1, 1)$

Now, $f(-1) = -1 + 3 - 1 = 1 > 0$

and $f(1) = 1 - 3 - 1 = -2 < 0$

Thus, the curve $y = f(x) = x^3 - 3x - 1$ will cut the x -axis exactly one point in $(-1, 1)$

25. Let $f(x) = e^x - 1 - x - \frac{x^2}{2}$

$$\Rightarrow f'(x) = e^x - 1 - x < 0, \forall x \in (-1, 1)$$

Thus, $f(x)$ is strictly decreases in $(-1, 1)$

Now, $f(-1) = \frac{1}{e} - 1 + 1 - \frac{1}{2} = \frac{1}{e} - \frac{1}{2} < 0$

and $f(1) = e - 1 - 1 - \frac{1}{2} = e - \frac{5}{2} > 0$

Thus, the equation $e^x = 1 + x + \frac{x^2}{2}$ has a real root in $(-1, 1)$

26. As we know that $\tan^{-1}x$ & e^x , both are strictly increasing for all x in R .

Therefore $f(x) = \tan^{-1}(e^x)$ is strictly increasing for all x in R .

27. As we know that $\tan^{-1}x$ is strictly increasing for all x in R and $(\log_{1/3}x)$ is strictly decreasing for all $x \in R^+$.

Therefore, $f(x) = \tan^{-1}(\log_{1/3}x)$ is strictly decreasing for all $x \in R$.

28. As we know that $\cot^{-1}x$ is strictly decreasing for all x in R and $(\log_4x)$ is increasing for all $x \in R^+$.

Therefore, $f(x) = \cot^{-1}(\log_4x)$ is strictly decreasing for all $x \in R^+$.

29. As we know that $(\log_{1/10}x)$ is strictly decreasing for all $x \in R^+$ and $(\cot^{-1}x)$ is strictly decreasing for all x .

Thus $f(x) = \cot^{-1}(\log_{1/10}x)$ is strictly increasing for all $x > 0$.

30. Let $f(x) = 3x - x^2$ and $g(x) = \sqrt{x}$

Now, $f'(x) = 3 - 2x$

By the sign scheme, f is strictly inc. in $\left(-\infty, \frac{3}{2}\right)$ and strictly decreasing in $\left(\frac{3}{2}, \infty\right)$.

Also, g is strictly increasing in $[0, \infty)$.

Now, $D_f = [0, 3]$

Thus, the function $y = \sqrt{3x - x^2}$ is strictly increasing in $\left(0, \frac{3}{2}\right)$ and strictly decreasing in $\left(\frac{3}{2}, 2\right)$.

31. Let $g(x) = \tan^{-1}$ and $h(x) = (\sin x + \cos x)$

Now, $h'(x) = (\cos x - \sin x)$

$$= \sqrt{2} \left(\frac{1}{\sqrt{2}} \cos x - \frac{1}{\sqrt{2}} \sin x \right)$$

$$= \sqrt{2} \cos \left(x + \frac{\pi}{4} \right)$$

$h(x)$ is strictly increasing if $h'(x) > 0$

$$\Rightarrow \sqrt{2} \cos \left(x + \frac{\pi}{4} \right) > 0$$

$$\Rightarrow \cos \left(x + \frac{\pi}{4} \right) > 0$$

$$\Rightarrow 0 < \left(x + \frac{\pi}{4} \right) < \frac{\pi}{2} \text{ \& } \frac{3\pi}{2} < \left(x + \frac{\pi}{4} \right) < 2\pi$$

$$\Rightarrow -\frac{\pi}{4} < x < \frac{\pi}{4} \text{ and } \frac{5\pi}{4} < x < \frac{7\pi}{4}$$

Thus, the given function $f(x)$ is strictly increases in

$$\left(\frac{\pi}{4}, \frac{3\pi}{4} \right) \cup \left(\frac{5\pi}{4}, \frac{7\pi}{4} \right) \text{ and strictly decreasing in } \left(0, \frac{\pi}{4} \right)$$

$$\cup \left(\frac{3\pi}{4}, \frac{5\pi}{4} \right) \cup \left(\frac{7\pi}{4}, 2\pi \right).$$

32. Let $g(x) = \frac{\log x}{x}$ and $h(x) = \log x$

Now, $g'(x) = \frac{1 - \log x}{x^2}$

By the sign scheme, $g(x)$ is strictly increasing in $(0, e)$ and strictly decreasing in (e, ∞)

Also, $h(x)$ is strictly increasing for all $x > 0$.

Also, $D_f = (1, \infty)$

Thus, $f(x)$ is strictly increasing in $(1, e)$

and strictly decreasing in (e, ∞)

33. Let $g(x) = \sin x + \cos x$ and $h(x) = \log x$

Since $h(x)$ is an increasing function, $f(x)$ will be decreases if $g(x)$ decreases.

Now, $g'(x) = \cos x - \sin x$

$$= \sqrt{2} \left(\frac{1}{\sqrt{2}} \cos x - \frac{1}{\sqrt{2}} \sin x \right)$$

$$= \sqrt{2} \cos \left(x + \frac{\pi}{4} \right)$$

Since $g(x)$ decreases, so $g'(x) < 0$

$$\Rightarrow \cos \left(x + \frac{\pi}{4} \right) < 0$$

$$\Rightarrow 2n\pi + \frac{\pi}{2} < \left(x + \frac{\pi}{4} \right) < 2n\pi + \frac{3\pi}{2}$$

$$\Rightarrow 2n\pi + \frac{\pi}{4} < x < 2n\pi + \frac{5\pi}{4}$$

$$\Rightarrow 2n\pi + \frac{\pi}{4} < \log x < 2n\pi + \frac{5\pi}{4}$$

$$\Rightarrow e^{\left(2n\pi + \frac{\pi}{4} \right)} < x < e^{\left(2n\pi + \frac{5\pi}{4} \right)}$$

34. Let $g(x) = \log_e x$ and $h(x) = \cos x$

Here, $g(x)$ is strictly increases for all $x > 0$

Also, $h(x)$ is strictly decreases in $(0, \pi)$

Again, for the domain of the function, $\cos x > 0$

$$\Rightarrow x \in \left(0, \frac{\pi}{2} \right)$$

Therefore, the function $f(x)$ is strictly decreases in

$$\left(0, \frac{\pi}{2} \right).$$

35. Let $g(x) = \sin x + \cos x$ and $h(x) = \sin x$

Now, $g'(x) = \cos x - \sin x$

$$= \sqrt{2} \cos \left(x + \frac{\pi}{4} \right)$$

When $g'(x) > 0$, $\sqrt{2} \cos \left(x + \frac{\pi}{4} \right) > 0$

$$\Rightarrow -\frac{\pi}{2} < \left(x + \frac{\pi}{4} \right) < \frac{\pi}{2}$$

$$\Rightarrow -\frac{\pi}{2} - \frac{\pi}{4} < x < \frac{\pi}{2} - \frac{\pi}{4}$$

$$\Rightarrow -\frac{3\pi}{4} < x < \frac{\pi}{4}$$

When $g'(x) < 0$, $\frac{\pi}{2} < \left(x + \frac{\pi}{4} \right) < \frac{3\pi}{2}$

$$\Rightarrow \frac{\pi}{2} - \frac{\pi}{4} < x < \frac{3\pi}{2} - \frac{\pi}{4}$$

$$\Rightarrow \frac{\pi}{4} < x < \frac{5\pi}{4}$$

Thus, $f(x)$ is strictly increases in $\left(0, \frac{\pi}{2} \right)$ and strictly

decreasing in $\left(\frac{\pi}{4}, \pi \right)$.

36. Let $f(x) = \log(1+x) - x + \frac{x^2}{2}$

$$\Rightarrow f'(x) = \frac{1}{1+x} - 1 + x$$

$$\Rightarrow f'(x) = \frac{1}{1+x} - (1-x)$$

$$\Rightarrow f'(x) = \frac{1 - 1 + x^2}{1+x} = \frac{x^2}{x+1} > 0 \forall x \in \mathbb{R}^+$$

$$\Rightarrow f(x) \text{ is strictly increasing in } (0, \infty)$$

$$\text{Thus } f(x) > f(0)$$

$$\Rightarrow \log(1+x) - x + \frac{x^2}{2} > 0$$

$$\Rightarrow \log(1+x) > x - \frac{x^2}{2}$$

Hence, the result.

$$37. \text{ Consider } f(x) = \log(1+x) - \frac{x}{x+1}$$

$$\Rightarrow f'(x) = \frac{1}{1+x} - \frac{1}{(x+1)^2}$$

$$\Rightarrow f'(x) = \frac{x+1-1}{(x+1)^2} = \frac{x}{(x+1)^2}$$

$$\Rightarrow f'(x) > 0 \text{ for all } x > 0$$

Thus, $f(x)$ is strictly increasing

$$\Rightarrow f(x) > f(0)$$

$$\Rightarrow \log(1+x) - \frac{x}{x+1} > 0$$

$$\Rightarrow \log(1+x) > \frac{x}{x+1} \text{ for all } x > 0$$

Hence, the result.

$$38. \text{ Let } f(x) = (1+x)\log(1+x) - e^x + 1$$

$$\Rightarrow f'(x) = \frac{(1+x)}{(1+x)} + \log(1+x) \cdot 1 - e^x$$

$$\Rightarrow f'(x) = 1 + \log(1+x) - e^x$$

$$\Rightarrow f'(x) < 0 \text{ for all } x < 0$$

Thus, $f(x)$ is strictly decreasing function

$$\Rightarrow f'(x) < f(0)$$

$$\Rightarrow (1+x)\log(1+x) - e^x + 1 < 0$$

$$\Rightarrow (1+x)\log(1+x) < e^x - 1$$

Hence, the result.

$$39. \text{ We have } f(x) = 2x \tan^{-1} x - \log(1+x^2)$$

$$\Rightarrow f'(x) = 2 \tan^{-1} x + \frac{2x}{1+x^2} - \frac{2x}{1+x^2}$$

$$\Rightarrow f'(x) = 2 \tan^{-1} x > 0 \text{ for all } x \text{ in } \mathbb{R}^+$$

Thus, $f(x)$ is strictly increasing in $\mathbb{R}^+$

$$\Rightarrow f(x) > f(0)$$

$$\Rightarrow 2x \tan^{-1} x - \log(1+x^2) > 0$$

$$\Rightarrow 2x \tan^{-1} x > \log(1+x^2)$$

Hence, the result.

$$40. \text{ Let } f(x) = 1 + x \log(x + \sqrt{x^2 + 1}) - \sqrt{1+x^2}$$

$$f'(x) = \log(x + \sqrt{x^2 + 1}) + \left(\frac{x}{x + \sqrt{x^2 + 1}} \right)$$

$$\left(1 + \frac{x}{\sqrt{x^2 + 1}} \right) - \frac{x}{\sqrt{1+x^2}}$$

$$f'(x) = \log(x + \sqrt{x^2 + 1}) + \frac{x}{\sqrt{1+x^2}}$$

$$- \frac{x}{\sqrt{1+x^2}}$$

$$f'(x) = \log(x + \sqrt{x^2 + 1})$$

$$f'(x) = \log(x + \sqrt{x^2 + 1}) \geq 0, \forall x \geq 0$$

Thus, $f(x)$ is increasing in $[0, \infty)$

$$f(x) \geq f(0)$$

$$1 + x \log(x + \sqrt{x^2 + 1}) - \sqrt{x^2 + 1} \geq 0$$

$$\Rightarrow 1 + x \log(x + \sqrt{x^2 + 1}) \geq \sqrt{x^2 + 1}$$

Hence, the result.

$$41. \text{ Let } f(x) = x - \sin x$$

$$\Rightarrow f'(x) = 1 - \cos x > 0, \forall x \in \left(0, \frac{\pi}{2}\right)$$

Thus, $f(x)$ is strictly increasing in $\left(0, \frac{\pi}{2}\right)$

$$\Rightarrow f(x) > f(0)$$

$$\Rightarrow x - \sin x > 0$$

$$\Rightarrow x > \sin x$$

$$\Rightarrow \cos x < \cos(\sin x) \quad \dots(i)$$

Also, for all x in $\left(0, \frac{\pi}{2}\right)$, $0 < \cos x < 1$

$$\Rightarrow \cos x < 1$$

$$\Rightarrow \cos x > \sin(\cos x) \quad \dots(ii)$$

From (i) and (ii), we get,

$$\sin(\cos x) < \cos x < \cos(\sin x)$$

$$42. \text{ Given } f(x) = \log x - Bx^2$$

$$\Rightarrow f'(x) = \frac{1}{x} - 2Bx = \frac{1 - 2Bx^2}{x}$$

The critical points of 'f' are

$$x = 0, \frac{1}{\sqrt{2B}}, -\frac{1}{\sqrt{2B}}$$

$$\text{Now, } f'(x) > 0, \forall x \in \left(0, \frac{1}{\sqrt{2B}}\right)$$

$$\text{and } f'(x) < 0, \forall x \in \left(\frac{1}{\sqrt{2B}}, \infty\right)$$

$$\text{Now, } \log x < Bx^2 \text{ for } x > 0$$

$$\text{It holds good for } x = \frac{1}{\sqrt{2B}}$$

$$\text{Thus } \log\left(\frac{1}{\sqrt{2B}}\right) < B, \frac{1}{2B} = \frac{1}{2}$$

$$\Rightarrow -\log(\sqrt{2B}) < \frac{1}{2}$$

$$\Rightarrow \log(\sqrt{2B}) < -\frac{1}{2}$$

$$\Rightarrow \sqrt{2B} < e^{-\frac{1}{2}}$$

$$\Rightarrow 2B < e^{-1}$$

$$\Rightarrow B > \frac{1}{2e}$$

$$\text{Thus, the least value of } B \text{ is } \frac{1}{2e}$$

$$43. \text{ Let } f(x) = ax^2 + \frac{b}{x} - c$$

$$\Rightarrow f'(x) = 2ax - \frac{b}{x^2}$$

$$\text{Now, } f'(x) = 0 \text{ gives } 2ax - \frac{b}{x^2} = 0$$

$$\Rightarrow 2ax^3 = b$$

$$\Rightarrow x = \left(\frac{b}{2a}\right)^{1/3}$$

$$\text{Thus, the least value of } f(x) \text{ occurs at } x = \left(\frac{b}{2a}\right)^{1/3}$$

$$\text{we have } a\left\{\frac{b}{2a}\right\}^{2/3} + \frac{b}{\left\{\frac{b}{2a}\right\}^{1/3}} \geq c$$

$$\Rightarrow a\left(\frac{b}{2a}\right) + b \geq c \cdot \left(\frac{b}{2a}\right)^{1/3}$$

$$\Rightarrow \left(\frac{3b}{2}\right)^3 \geq \frac{b}{2a} \cdot c^2$$

$$\Rightarrow \frac{27b^3}{8} \geq \frac{b}{2a} \cdot c^2$$

$$\Rightarrow 27b^2 a \geq 4c^3$$

$$\Rightarrow 27ab^2 \geq 4c^3$$

Hence, the result.

$$44. \text{ We have } f(x) = x^5 + 5x - 6$$

$$\Rightarrow f'(x) = 5x^4 + 5$$

$$\Rightarrow f''(x) = 20x^3$$

$$\Rightarrow f'''(x) = 60x^2 \text{ exists for all } x$$

$$\text{Now, } f''(x) = 0 \text{ gives } x = 0$$

By the sign scheme for $f''(x) = 0$, we have, $f(x)$ is concave down in $(-\infty, 0)$ and concave up in $(0, \infty)$.

$$45. \text{ We have } f(x) = x^4 - 5x^3 - 15x^2 + 30$$

$$\Rightarrow f'(x) = 4x^3 - 15x^2 - 30x$$

$$\Rightarrow f''(x) = 12x^2 - 30x - 30$$

$$\text{Now, } f''(x) = 0 \text{ gives } 12x^2 - 30x - 30 = 0$$

$$\Rightarrow 6x^2 - 15x - 15 = 0$$

$$\Rightarrow 2x^2 - 5x - 5 = 0$$

$$\Rightarrow x = \frac{5 \pm \sqrt{25 + 40}}{4}$$

$$\Rightarrow x = \frac{5 \pm 8}{4} = \frac{13}{4}, -\frac{3}{4}$$

By the sign scheme for the function $f''(x)$, the function $f(x)$ is concave down in $\left(-\frac{3}{4}, \frac{13}{4}\right)$ and concave up in $\left(-\infty, -\frac{3}{4}\right) \cup \left(\frac{13}{4}, \infty\right)$

$$46. \text{ We have } f(x) = (\sin x + \cos x)e^x$$

$$f'(x) = (\sin x + \cos x)e^x + e^x(\cos x - \sin x)$$

$$f'(x) = e^x(\sin x + \cos x + \cos x - \sin x)$$

$$f'(x) = 2e^x \cos x$$

$$f''(x) = 2(e^x \cos x - e^x \sin x)$$

$$f''(x) = 2e^x(\cos x - \sin x)$$

$$\text{Now, } f''(x) = 0 \text{ gives } 2e^x(\cos x - \sin x) = 0$$

$$\Rightarrow \tan x = 1$$

$$\Rightarrow x = \frac{\pi}{4}, \frac{5\pi}{4}$$

By the sign scheme for the function $f''(x) = 0$,

have $f(x)$ is concave down in $\left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$ and

up in $\left(0, \frac{\pi}{4}\right) \cup \left(\frac{5\pi}{4}, 2\pi\right)$

$$47. \text{ Given curve is } y = f(x) = Ax^2 + Bx + C$$

$$f'(x) = 2Ax + B$$

$$f''(x) = 2A$$

Thus, the curve concave up if $f''(x) > 0$ and concave down if $f''(x) < 0$ i.e. concave up if $A > 0$: concave down if $A < 0$.

48. We have $f(x) = x^4 - 4x^3 + x - 10$

$$\Rightarrow f'(x) = 4x^3 - 12x^2 + 1$$

$$\Rightarrow f''(x) = 12x^2 - 24x = 12x(x - 2)$$

Now, $f''(x) = 0$ gives $x = 0$ and $x = 2$.

when $x = 0, y = -10$

when $x = 2, y = -24$

Thus, the point of inflection are $(0, -10)$ and $(2, -24)$

49. We have $y = f(x) = (x - 2)^{2/3} + 10$

$$\Rightarrow f'(x) = \frac{2}{3}(x - 2)^{-1/3} = \frac{2}{3(x - 2)^{1/3}}$$

$$\Rightarrow f''(x) = \frac{2}{9(x - 2)^{4/3}}$$

Thus, $f''(x)$ does not exist at $x = 2$.

when $x = 2, y = 0 + 10 = 10$

Thus, the inflection point is $(2, 10)$

50. We have $f(x) = x^4 - 6x^3 + 12x^2 - 8x + 3$

$$\Rightarrow f'(x) = 4x^3 - 18x^2 + 24x - 8$$

$$\begin{aligned}\Rightarrow f''(x) &= 12x^2 - 36x + 24 \\ &= 12(x^2 - 3x + 2) \\ &= 12(x - 1)(x - 2)\end{aligned}$$

Now, $f''(x) = 0$ gives $x = 1, 2$

when $x = 1, y = 2$

when $x = 2, y = 3$

Thus, the point of inflection are $(1, 2)$ and $(2, 3)$.

51. Given $y = f(x) = x^2 - \frac{1}{6x^3}$

$$\Rightarrow f'(x) = 2x + \frac{1}{2x^4}$$

$$\Rightarrow f''(x) = 2 - \frac{2}{x^5}$$

Now, $f''(x) = 0$ gives $2 - \frac{2}{x^5} = 0$

Thus, $x = 1$

when $x = 1, y = \frac{5}{6}$

Thus, the point of inflection is $\left(1, \frac{5}{6}\right)$.

52. We have $y = f(x) = e^{-x^2}$

$$\Rightarrow f'(x) = e^{-x^2} \times -2x$$

$$\Rightarrow f''(x) = -2e^{-x^2} \times x$$

$$\Rightarrow f'''(x) = -2(e^{-x^2} \cdot 1 + x \cdot e^{-x^2} \cdot (-2x))$$

$$\Rightarrow f'''(x) = 2e^{-x^2}(2x^2 - 1)$$

Now, $f'''(x) = 0$ gives $2e^{-x^2}(2x^2 - 1) = 0$

$$\Rightarrow x = \pm \frac{1}{\sqrt{2}}$$

when $x = \frac{1}{\sqrt{2}}, y = e^{-\frac{1}{2}} = \frac{1}{\sqrt{e}}$

when $x = -\frac{1}{\sqrt{2}}, y = e^{-\frac{1}{2}} = \frac{1}{\sqrt{e}}$

Thus, the point of inflection are

$$\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{e}}\right) \& \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{e}}\right)$$